

A sharp one-sided real-axis bound for the fast exponential Schrodinger example

Run `fa_banach_001`, agent `agent_lane_00`

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Status

This is a partial-result packet. It proves the lower-bound, or quasimode, side of the real-axis asymptotic predicted for the fast exponential example in Arnal–Siegl. It does not prove the matching upper bound, and therefore does not close the source open question.

Source and Open Question

The source is Antonio Arnal and Petr Siegl, “Resolvent estimates for one-dimensional Schrodinger operators with complex potentials”, arXiv:2203.15938, published in *Journal of Functional Analysis* 284 (2023), Article 109856 [1]. Section 7.3 considers

$$H = -\partial_x^2 + ie^{x^2}, \quad \text{Dom}(H) = W^{2,2}(\mathbb{R}) \cap \text{Dom}(e^{x^2}),$$

and gives the imaginary-axis asymptotic. It then states that the real-axis Theorem 4.2 is not applicable and leaves the critical region next to the real axis open.

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7.3. Fast growing potential. Let $H = -\partial_x^2 + ie^{x^2}$ with $\text{Dom}(H) = W^{2,2}(\mathbb{R}) \cap \text{Dom}(e^{x^2})$. Then

$$\Psi(ib) = 2^{-\frac{2}{3}} \|A_{1, \frac{\pi}{2}}^{-1}\| b^{-\frac{2}{3}} (\log b)^{-\frac{1}{3}} (1 + \mathcal{O}(b^{-\frac{1}{3}} (\log b)^{\frac{1}{3}})), \quad b \rightarrow +\infty, \quad (7.6)$$

which is as before illustrated in Fig. 7.1. Since the decay of $\Psi(\lambda)$ on the imaginary axis is faster than for any polynomial potential, the region for uniform boundedness of $\Psi(\lambda)$ adjacent to the imaginary axis is correspondingly wider. Note that Theorem 4.2 on the behaviour of $\Psi(\lambda)$ for $\lambda \in \mathbb{R}_+$ is not applicable in this case, see also Figure 7.1, and therefore the description of the critical region next to the real axis is currently an open question although [30, Eq. (5.5)] provides a clue as to what it may look like.

APPENDIX A GENERALISED AIRY OPERATOR

Transcription of the Relevant Source Point

The relevant paragraph in Section 7.3 states that for $H = -\partial_x^2 + ie^{x^2}$ the real-axis theorem from the paper is not applicable, so the critical region next to the positive real axis remains open. This packet addresses one direction of that missing real-axis analysis.

Proof Intuition

Near the positive turning frequency $\xi = \sqrt{a}$, a wave packet of the form

$$u(x) = t_a^{-1/2} e^{i\sqrt{a}x} v(x/t_a), \quad t_a e^{t_a^2} = 2\sqrt{a},$$

turns the second-order operator $H - a$ into a first-order transport operator:

$$(H - a)u = ie^{t_a^2} t_a^{-1/2} e^{i\sqrt{a}x} \left(-v' + e^{t_a^2(y^2-1)} v \right)_{y=x/t_a} + \text{a smaller second-derivative term.}$$

The normalized first-order model

$$B_t = -\partial_y + e^{t^2(y^2-1)}$$

converges to Volterra integration on $(-1, 1)$, whose norm is $4/\pi$. Thus choosing v from a near-extremizer of B_t^{-1} produces physical quasimodes for $H - a$ with residual at most $(\pi/4 + o(1))e^{t_a^2}$. This proves the sharp predicted lower bound for the resolvent norm. What remains open is the converse inequality for arbitrary test functions.

Definitions

For $t > 0$, define

$$W_t(y) := e^{t^2(y^2-1)}, \quad B_t = -\partial_y + W_t$$

on $L^2(\mathbb{R})$ with maximal domain

$$\text{Dom}(B_t) = \{v \in L^2(\mathbb{R}) : -v' + W_t v \in L^2(\mathbb{R})\}.$$

Let T_0 be the limiting Volterra operator

$$(T_0 f)(x) = \int_x^1 f(y) dy, \quad -1 < x < 1,$$

viewed as an operator on $L^2(-1, 1)$. Its norm is $4/\pi$.

Theorem

Theorem 1 (sharp one-sided real-axis bound). *Let*

$$H = -\partial_x^2 + ie^{x^2}, \quad \text{Dom}(H) = W^{2,2}(\mathbb{R}) \cap \text{Dom}(e^{x^2}),$$

and let $t_a > 0$ be determined by

$$t_a e^{t_a^2} = 2\sqrt{a}.$$

Then

$$\liminf_{a \rightarrow +\infty} e^{t_a^2} \|(H - a)^{-1}\| \geq \frac{4}{\pi}.$$

Equivalently, if

$$m(a) := \inf_{0 \neq u \in \text{Dom}(H)} \frac{\|(H - a)u\|}{\|u\|},$$

then

$$\limsup_{a \rightarrow +\infty} e^{-t_a^2} m(a) \leq \frac{\pi}{4}.$$

Proof

Lemma 1 (model inverse convergence). *Let $T_t = B_t^{-1}$. For every $f \in C_c^\infty((-1, 1))$,*

$$T_t f \longrightarrow T_0 f \quad \text{in } L^2(\mathbb{R}), \quad t \rightarrow +\infty,$$

where $T_0 f$ is extended by zero outside $(-1, 1)$ on the right and by its Volterra value on $(-1, 1)$.

Proof. Solving the first-order equation with the L^2 condition at $+\infty$ gives

$$(T_t f)(x) = \int_x^\infty \exp\left(-\int_x^y W_t(s) ds\right) f(y) dy.$$

For $-1 < x < 1$, $W_t \rightarrow 0$ locally uniformly, and for intervals outside $[-1, 1]$, $W_t \rightarrow +\infty$ locally uniformly. Hence the kernel converges pointwise to $\mathbf{1}_{\{-1 < x < y < 1\}}$. The same tail estimates as in the standard proof of the fast-model packet give convergence in Hilbert–Schmidt norm on the support of f , and therefore $T_t f \rightarrow T_0 f$ in L^2 . $\square$

Lemma 2 (smooth near-extremizers). *For every $\varepsilon > 0$, there is $f \in C_c^\infty((-1, 1))$ such that*

$$\frac{\|f\|_{L^2(-1,1)}}{\|T_0 f\|_{L^2(-1,1)}} \leq \frac{\pi}{4} + \varepsilon.$$

Proof. The Volterra operator on an interval of length 2 has norm $4/\pi$. Choose an L^2 near-extremizer for T_0 , then approximate it in L^2 by a smooth compactly supported function. Since T_0 is bounded, the ratio is preserved up to an arbitrarily small error. $\square$

Lemma 3 (second derivative of the first-order quasimode). *Fix $f \in C_c^\infty((-1, 1))$, and put $v_t := T_t f$. Then there is a constant C_f such that, as $t \rightarrow +\infty$,*

$$\|v_t''\|_{L^2(\mathbb{R})} \leq C_f t^5.$$

Proof. Let $\text{supp } f \subset [\alpha, \beta]$, with $-1 < \alpha < \beta < 1$. The identity $B_t v_t = f$ gives

$$v_t' = W_t v_t - f, \quad v_t'' = (W_t' + W_t^2) v_t - W_t f - f'.$$

On $[\alpha, \beta]$, the functions W_t and W_t' are uniformly bounded, and outside $(-\infty, \beta]$ we have $v_t = 0$. On $[-1, \alpha]$, $f = 0$, $0 \leq W_t \leq 1$, and $|W_t'| \leq 2t^2 W_t$, while $\|v_t\|_\infty \leq \|f\|_1$. These pieces contribute $O_f(t^2)$ to the L^2 -norm of v_t'' .

It remains to consider $x < -1$. There $f = 0$, so

$$v_t(x) = C_t \exp\left(-\int_x^\alpha W_t(s) ds\right), \quad |C_t| \leq \|f\|_1.$$

Write $x = -1 - r$, $r > 0$. On $0 < r < 1$, set $z = e^{2t^2 r}$. Since

$$W_t(-1 - r) \leq z^{3/2}, \quad \int_{-1-r}^{-1} W_t(s) ds \geq \frac{z-1}{2t^2},$$

we get

$$\int_{-2}^{-1} |v_t''(x)|^2 dx \leq C_f \int_1^\infty (t^4 z^3 + z^6) \exp\left(-\frac{z-1}{t^2}\right) \frac{dz}{t^2 z} \leq C_f t^{10}.$$

For $x < -2$, the integral $\int_x^{-1} W_t(s) ds$ grows faster than any fixed exponential in $|x|$, while W_t and W_t' grow only like exponentials in x^2 ; the same estimate on unit intervals gives a bounded contribution. Hence $\|v_t''\|_2 = O_f(t^5)$. $\square$

Proof of Theorem 1. Fix $\varepsilon > 0$, and choose $f \in C_c^\infty((-1, 1))$ as in Lemma 2. Let $v_t = T_t f$. For $a > 0$, put

$$t = t_a, \quad \xi = \sqrt{a}, \quad u_a(x) = t^{-1/2} e^{i\xi x} v_t(x/t).$$

Then $\|u_a\|_{L^2(\mathbb{R})} = \|v_t\|_{L^2(\mathbb{R})}$, and a direct calculation gives, with $y = x/t$,

$$(H - a)u_a(x) = t^{-1/2} e^{i\xi x} \left(-t^{-2} v_t''(y) - 2i\xi t^{-1} v_t'(y) + i e^{t^2 y^2} v_t(y) \right).$$

Since $te^{t^2} = 2\xi$, this becomes

$$(H - a)u_a = i e^{t^2} t^{-1/2} e^{i\xi x} (B_t v_t)(x/t) - t^{-5/2} e^{i\xi x} v_t''(x/t).$$

Taking norms and using $B_t v_t = f$, Lemma 3, and Lemma 1,

$$e^{-t^2} \frac{\|(H - a)u_a\|}{\|u_a\|} \leq \frac{\|f\| + t^{-2} e^{-t^2} \|v_t''\|}{\|v_t\|} \longrightarrow \frac{\|f\|}{\|T_0 f\|} \leq \frac{\pi}{4} + \varepsilon.$$

Therefore

$$\limsup_{a \rightarrow +\infty} e^{-t_a^2} m(a) \leq \frac{\pi}{4} + \varepsilon.$$

Letting $\varepsilon \downarrow 0$ proves the singular-value statement, and inverting it gives

$$\liminf_{a \rightarrow +\infty} e^{t_a^2} \|(H - a)^{-1}\| \geq \frac{4}{\pi}.$$

$\square$

Verification Notes

No computation is used in the proof. The optional script `code/quasimode_residual_check.py` evaluates the scaling identity on a grid for moderate t , using a compactly supported smooth input f . It is only a sanity check for the algebraic rescaling and the decay of the second-derivative correction.

Limitations and Novelty Check

This result does not prove the matching upper bound

$$\|(H - a)^{-1}\| \leq \left(\frac{4}{\pi} + o(1)\right) e^{-t_a^2}.$$

That upper bound would require a global estimate for arbitrary u , replacing the Fourier partition and commutator estimates used in Arnal–Siegl Theorem 4.2. Those estimates rely on regular variation and symbol-class derivative bounds, while $V_2(x) = e^{x^2}$ has $V_2^{(n)}(x) = P_n(x)e^{x^2}$ and is not regularly varying.

The run’s cheap indexes were checked on 2026-07-18 for arXiv:2203.15938 and the fast-exponential real-axis keywords. They contained the earlier conditional model packet but no full or partial packet proving this one-sided source-operator bound. The companion generalized Airy paper was also checked; it records applications to the regularly varying real-axis theorem and does not supply the missing fast-exponential real-axis transfer.

Human Review Recommendation

Review as a likely valid partial result. The most important proof checkpoint is Lemma 3: it must be enough that $\|v_t''\|$ grows only polynomially, because the rescaling multiplies this term by $t^{-2}e^{-t^2}$. The packet should not be treated as closing the open critical-region question until the converse, all-functions upper bound is proved.

References

- [1] Antonio Arnal and Petr Siegl, *Resolvent estimates for one-dimensional Schrodinger operators with complex potentials*, Journal of Functional Analysis 284 (2023), no. 9, Article 109856; arXiv:2203.15938.